

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEIGN AND CONNECTIVITY

Course Code:	DSA05	Course Title:	Query Processing for Data Science With Real Time Applications
CO Assessed:	CO2	Assessment:	Tool 1 – DEIGN AND CONNECTIVITY
Weightage:		Total Marks:	40 Marks
CO2 Statement: Use Python basics and SQL libraries to connect to databases SQLite, MySQL, PostgreSQL and perform CRUD operations like Create, Read, Update, Delete. (BTL6)			
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Section / Batch:	D /2024	Date of Submission:	05/08/26
Faculty In-charge:	Dr. B.SHYAMALA GOWRI	Project Mode:	Individual Submission

Code of Conduct

I _BHARANI DHARAN S(192424141) (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: ____BHARANI S_____

Requirement Analysis (5 Marks)	ER Diagram Design (10 Marks)	Table Design & Normalization (5 Marks)	Database Connectivity using Python (20 Marks)	CRUD Operations (5 Marks)	Total (35 Marks)

Library Management System Database Design

1. Database Tables

Books Table

Column Name	Data Type	Description
book_id	INTEGER	Primary Key
title	TEXT	Book name
author_id	INTEGER	Foreign Key
publisher_id	INTEGER	Foreign Key
category	TEXT	Book category
quantity	INTEGER	Number of copies

Authors Table

Column Name	Data Type	Description
author_id	INTEGER	Primary Key
author_name	TEXT	Author name

Publishers Table

Column Name	Data Type	Description
publisher_id	INTEGER	Primary Key
publisher_name	TEXT	Publisher name

Members Table

Column Name	Data Type	Description
member_id	INTEGER	Primary Key
member_name	TEXT	Member name
department	TEXT	Department
phone	TEXT	Contact number

Transactions Table

Column Name	Data Type	Description
transaction_id	INTEGER	Primary Key
member_id	INTEGER	Foreign Key
book_id	INTEGER	Foreign Key
issue_date	DATE	Issue date
due_date	DATE	Due date
return_date	DATE	Return date
fine	REAL	Fine amount

Entity Relationship (ER) Diagram

- One author → many books.
- One publisher → many books.
- One member → many transactions.
- One book → many transactions.

Python Program (SQLite)

```
import sqlite3
```

```
from datetime import datetime
```

```
# Connect to database
```

```
conn = sqlite3.connect("library.db")
```

```
cursor = conn.cursor()
```

```
# Create tables
```

```
cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS authors (
```

```
author_id INTEGER PRIMARY KEY,  
author_name TEXT  
)  
""")
```

```
cursor.execute("""  
CREATE TABLE IF NOT EXISTS publishers (  
    publisher_id INTEGER PRIMARY KEY,  
    publisher_name TEXT  
)  
""")
```

```
cursor.execute("""  
CREATE TABLE IF NOT EXISTS books (  
    book_id INTEGER PRIMARY KEY,  
    title TEXT,  
    author_id INTEGER,  
    publisher_id INTEGER,  
    category TEXT,  
    quantity INTEGER,  
    FOREIGN KEY(author_id) REFERENCES authors(author_id),  
    FOREIGN KEY(publisher_id) REFERENCES publishers(publisher_id)  
)  
""")
```

```
cursor.execute("""  
CREATE TABLE IF NOT EXISTS members (  
    member_id INTEGER PRIMARY KEY,  
    member_name TEXT,  
    department TEXT,  
    phone TEXT  
)  
""")
```

```
cursor.execute("""  
CREATE TABLE IF NOT EXISTS transactions (  
    transaction_id INTEGER PRIMARY KEY,  
    member_id INTEGER,  
    book_id INTEGER,  
    issue_date TEXT,  
    due_date TEXT,  
    return_date TEXT,  
    fine REAL,  
    FOREIGN KEY(member_id) REFERENCES members(member_id),  
    FOREIGN KEY(book_id) REFERENCES books(book_id)  
)  
""")
```

```
conn.commit()
```

```
# Insert sample data
```

```
cursor.execute(  
    "INSERT INTO authors VALUES (1, 'R.K. Narayan')"  
)
```

```
cursor.execute(  
    "INSERT INTO publishers VALUES (1, 'Oxford')"  
)
```

```
cursor.execute(  
    "INSERT INTO books VALUES (1, 'Malgudi Days', 1, 1, 'Fiction', 10)"  
)
```

```
cursor.execute(  
    "INSERT INTO members VALUES (1, 'Arun', 'IT', '9876543210')"  
)
```

)

conn.commit()

Issue book

issue_date = "2026-08-01"

due_date = "2026-08-15"

cursor.execute("""

INSERT INTO transactions

(member_id, book_id, issue_date, due_date, return_date, fine)

VALUES (?, ?, ?, ?, ?, ?)

""", (1, 1, issue_date, due_date, None, 0))

conn.commit()

Calculate fine

return_date = datetime.strptime("2026-08-20", "%Y-%m-%d")

due_date_obj = datetime.strptime(due_date, "%Y-%m-%d")

days_late = (return_date - due_date_obj).days

fine = days_late * 5 if days_late > 0 else 0

print("Fine amount =", fine)

conn.close()